

PROJECT DOCUMENTATION

SMARTPHARMA

Wholesale Pharmacy Management System

SAP ABAP 7.4

Main Program: ZSMARTPHARMA

Transaction Code: **ZSP**

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1. Introduction

SmartPharma – Wholesale Pharmacy Management System is a custom SAP ABAP 7.4 application developed to support the main activities of a wholesale medical store. The application provides a menu-driven SAP GUI interface for customer management, vendor management, purchase processing, sales and billing, stock monitoring, and reporting.

The project uses custom Z tables to store pharmacy-specific master and transaction data. ALV is used to present report output, while Smart Forms are used for invoice generation. The application is implemented as an ABAP backend solution without a separate external frontend.

2. Project Overview

The main executable program is ZSMARTPHARMA and the requested transaction code is ZSP. The application starts with a main selection screen from which the user can select a business module. Each module provides its relevant display, create, history, details, or billing operation.

The project follows a simple flow: select a module, enter or select the required operation, validate the input, access the corresponding custom table data, process the transaction, and display the result through ALV or invoice output.

3. Objectives

- Maintain wholesale pharmacy customer and vendor information.
- Record purchase transactions and purchase items.
- Record sales transactions and billing information.
- Maintain medicine stock by material and batch.
- Track purchase price, sales price, quantity and expiry date.
- Avoid duplicate customer records by checking the customer mobile number during billing.
- Display application data through standard SAP ALV functionality.
- Generate a customer bill through Smart Form ZSF_SMARTPHARMA_BILL.

4. Scope of the Project

The project covers the backend business processes required for a small wholesale pharmacy management application inside SAP. The implemented scope includes custom database tables, selection-screen navigation, master data maintenance, purchase and sales data processing, stock updates, ALV reports, and Smart Form invoice processing.

The project does not use a separate web or mobile frontend. User interaction is performed through SAP GUI selection screens and ALV output.

5. Technology and Tools

Technology / Tool	Usage
SAP ABAP 7.4	Application development
SAP GUI	Application execution and testing
Eclipse / ABAP Development Tools	ABAP development environment
SE11 / DDIC	Custom table and data dictionary objects
CL_SALV_TABLE	ALV report display
Smart Forms	Invoice generation
Custom Z Tables	Pharmacy master and transaction storage

6. System Architecture

The application follows a modular ABAP architecture in which the main program controls navigation and individual processing routines handle the business operations.

Main flow:

ZSP → ZSMARTPHARMA → Main Menu → Module Selection → Business Processing → Custom Z Tables → ALV / Smart Form Output

Major application modules:

- Customer Management
- Vendor Management
- Purchase Management
- Sales and Billing
- Stock Management
- Stock Dashboard / Reporting

7. Database Design

The project uses custom Z tables so that pharmacy-specific information can be stored independently of unrelated standard application data.

Object	Purpose
ZSP_CUSTOMER	To store customer details.
ZSP_VENDOR	To store vendor details.
ZSP_PURCHASE_HDR	To store purchase details.
ZSP_PURCH_ITEM	To store purchase medicine/item information.
ZSP_SALES_HDR	To store sales details.
ZSP_SALES_ITEM	To store sales medicine/item information.
ZSP_INV_STOCK	To store inventory & stock details

Key relationships

- Customer information is referenced by sales/billing using the customer ID.
- Vendor information is referenced by purchase header using the vendor ID.
- Purchase header and purchase item are linked through purchase ID.
- Sales header and sales item are linked through sales ID.
- Inventory is identified using material number and batch number.
- Sales item processing reduces the corresponding stock quantity.
- Sales item information is mapped to the Smart Form bill-item structure for invoice output.

8. Module Description

The main selection screen provides radio-button based navigation to the application modules. Each module opens its own operation menu so that the user can select the required business activity.

9. Customer Management

The Customer module provides customer display, customer history and customer creation functions. Customer information is stored in ZSP_CUSTOMER.

During billing, the customer mobile number is checked against existing customer records. If a matching customer is found, the existing customer information is reused. This supports repeat visits without creating duplicate customer master records.

10. Vendor Management

The Vendor module maintains wholesale supplier information. Vendor records include vendor ID, vendor name, contact person, mobile number, GST number, address and creation date. The vendor ID is used by purchase processing.

11. Purchase Management

Purchase processing stores information at two levels. ZSP_PURCHASE_HDR stores the purchase header, while ZSP_PURCH_ITEM stores the medicine/item details. Purchase item data includes material number, batch number, quantity, unit, purchase price and total amount.

The purchase module provides display, detail and create operations through the application's selection-screen navigation.

12. Sales and Billing

The Sales module provides sales display, sales details and billing functionality. Billing begins by requesting the customer's mobile number. Existing customers can be reused, while a new customer can be created when no matching mobile number is found.

Medicine entries include material number, batch number and quantity. The system checks stock availability, calculates item amounts and the overall bill amount, creates a sales header and sales items, and updates stock quantities.

The billing process uses the sales header and sales item tables to maintain transaction history.

13. Stock Management

Stock is maintained in ZSP_INV_STOCK using material number and batch number as the key identification. The table stores available quantity, unit, purchase price, sales price and expiry date.

The Stock module provides views for all stock, low stock, near-expiry stock and expired stock. These views help the user monitor inventory conditions.

14. ALV Reporting

Application output is displayed using CL_SALV_TABLE. Standard SALV functions provide common ALV operations such as sorting, filtering and output handling. Custom toolbar functions were not required for the implemented solution.

The ALV reports are intended to provide readable views of customer, vendor, purchase, sales and stock information.

15. Smart Form Invoice

The project includes Smart Form ZSF_SMARTPHARMA_BILL for invoice generation. Standard Smart Form control and output parameters are used together with custom billing parameters and a billing item internal table.

Custom invoice information includes bill number, date, customer ID, customer name, mobile number, GST number, address and total amount. The item data is represented using ZST_SP_BILL_ITEM and ZTT_SP_BILL_ITEM.

The intended billing flow is:

Billing → Save Sales Data → Build Bill Item Data → Call Smart Form → Invoice Preview / Output

16. Program Flow

1. Start ZSMARTPHARMA.
2. Display the main menu.
3. Select Customer, Vendor, Purchase, Sales, Stock or Dashboard.
4. Display the module-specific operation screen.

5. Execute the selected operation.
6. Validate user input.
7. Read or update the corresponding custom Z tables.
8. Display report results using ALV where applicable.
9. For billing, save sales information, update stock and call the Smart Form.
10. Complete the transaction and return control according to the application's navigation flow.

17. Validation and Error Handling

- Required billing customer mobile number is validated.
- Existing customers are searched using mobile number before creating a new customer.
- Medicine and batch information is checked against inventory.
- Requested quantity is checked against available stock.
- Optional medicine entries are processed only when supplied.
- Total amounts are calculated before the transaction is committed.
- Database changes are committed after successful transaction processing.
- Messages are displayed when required information is missing, stock is unavailable, or a requested record cannot be found.

18. Testing

Testing of the application should be performed module by module using representative master and transaction data.

Test Area	Expected Result
Customer Create	A valid customer record is saved in ZSP_CUSTOMER.
Customer Repeat Billing	Existing customer is reused when the mobile number matches.
Vendor Create	A valid vendor record is saved in ZSP_VENDOR.
Purchase Create	Purchase header and item data are stored.
Sales/Billing	Sales header and item records are created and total is calculated.
Stock Update	Stock quantity is reduced according to the billed quantity.
Stock Reports	Relevant stock records are displayed through ALV.
Smart Form	Billing information is passed to ZSF_SMARTPHARMA_BILL for invoice output.

19. Advantages and Limitations

Advantages

- Centralized pharmacy master and transaction information.
- Simple SAP GUI based menu-driven operation.
- Custom Z tables provide project-specific data storage.
- ALV provides standard report functions.
- Batch and expiry information is maintained with stock.
- Repeat customers can be recognized using mobile number.
- Billing supports invoice generation through Smart Forms.

Limitations

- The solution is an ABAP backend/SAP GUI application and does not provide a separate web or mobile frontend.

- The current billing entry screen is designed around the implemented medicine-entry fields rather than a dynamic item-entry grid.
- The application depends on the configured SAP development environment, authorizations and custom DDIC objects.

20. Future Enhancements

- Add a fully dynamic item-entry screen for any number of medicines per bill.
- Improve invoice layout with a complete medicine-item table and additional pharmacy branding.
- Add automated stock alerts and expiry notifications.
- Add user authorization by role for cashier, purchase and administration functions.
- Add richer dashboard charts and business KPIs.
- Add more detailed customer purchase-history reporting.
- Integrate barcode scanning and medicine master data where appropriate.

21. Conclusion

SmartPharma – Wholesale Pharmacy Management System demonstrates how SAP ABAP can be used to develop a complete domain-specific business application using custom DDIC tables, selection screens, modular processing, ALV reporting and Smart Forms. The project covers the core data and transaction flow required for wholesale pharmacy management, from customer and vendor maintenance through purchasing, stock control, sales and billing.

The application provides a structured foundation that can be extended with additional automation, reporting, authorization and user-interface capabilities.

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